

Samuel Berrios, Jr., MBA, PMP

PROFILE

- Cybersecurity leader with a proven track record of strengthening the confidentiality, integrity, availability, authenticity, and non-repudiation of enterprise systems by leading mission-critical initiatives that minimize systemic information security risk and maximize network resilience.
- Proficient in defensive cyber operations, web application & cloud security, vulnerability management, incident response, cyber threat intelligence, enterprise process automation, and counter-UAS (C-UAS) threat detection and mitigation.
- Expert in mission-critical process reengineering for enterprise cybersecurity, technology, and data ecosystems thru baselining, redesigning, and realigning core systems to strengthen security architectures, streamline operations, and accelerate digital transformation.
- Adept at translating complex internal and external data and AI-driven analytics into clear, actionable insights that drive informed decision-making and sound policy outcomes.
- Engages diverse stakeholders at every organizational level to navigate critical challenges, surface emerging opportunities, and simplify systemic complexity using the same collaborative discipline that transforms mission-critical information security initiatives into quantifiable, sustainable, lasting strategic impact.

EDUCATION

University of Texas at San Antonio - Carlos Alvarez College of Business, San Antonio, TX

- University of Texas (UT) System Online Consortium Master of Business Administration (MBA) - 2004
- Bachelor of Business Administration (BBA), Accounting - 2001
- Bachelor of Business Administration (BBA), Finance - 2001
- Bachelor of Business Administration (BBA), Management, International Business Concentration - 2000

Leadership: Founding President, Undergraduate Business Student Council (BSC); Founding President, Phi Chi Theta National Coed Professional and Economics Fraternity, Epsilon Omega Chapter; Founding Member, Delta Sigma Phi Social Fraternity, Theta Phi Chapter

Honors: Beta Alpha Psi International Honor Organization for Financial Information Students and Professionals; Mortar Board National College Senior Honor Society; Omicron Delta Epsilon International Honor Society in Economics; Omicron Delta Kappa National Leadership Honor Society

AREAS OF EXPERTISE

Vulnerability and Continuous Threat Exposure Management · Cyber Incident Response · Defensive Cyber Operations · Information Technology Project Portfolio Management · Agentive/Robotic/Business Process Automation · Ethical Hacking · Cyber Threat Intelligence · Information Security Strategy and Policy Management · Data Analytics

PROFESSIONAL APPOINTMENTS

Consumer Financial Protection Bureau (CFPB), Washington, DC

Jan 2017 – Present

Cybersecurity Operations Specialist

Jul 2022 – Present

Strengthen Bureau cybersecurity posture by leading technical and operational projects for the Chief Information Security Officer (CISO) Cybersecurity Operations (CyberOps) to reduce system vulnerabilities and bolster network resilience. Hold Position of Public Trust (PPT).

- Reduce agency cyber risk through continuous threat exposure management (CTEM) approach securing attack surface by scoping and discovering infrastructure exposures via vulnerability scanning (Qualys) and endpoint visibility (CrowdStrike), prioritizing findings by exploitability, impact, and attack vector using cyber threat intelligence (Anomali ThreatStream), standard frameworks (MITRE ATT&CK/CAPEC, CVE/CWE, FIRST CVSS/EPSS), and structured decision models (CISA SSVC, FAIR), validating results through breach and attack simulation (BAS) and internal & external penetration testing, and mobilizing remediation teams to close findings (patching, software updates, configuration changes) through ServiceNow.
- Lead cross-functional IT teams on Agile/SDLC cybersecurity implementation projects for CyberOps, including the Department of Justice (DOJ) Security Operations Center-as-a-Service (SOCaaS) platform transition and Cybersecurity and Infrastructure Security (CISA) Secure Cloud Business Applications (SCuBA) implementation, and oversee external penetration testing engagements from scoping through remediation to identify attack vectors and drive risk mitigation.
- Lead development and lifecycle management of cybersecurity governance playbooks (e.g., incident response and recovery and vulnerability management), driving standardization and compliance with federal and Bureau IT requirements.
- Direct cybersecurity procurement and financial management, including market research, proposal evaluation, Federal Risk and Authorization Management Program (FedRAMP) compliance, contractor performance oversight, and budget planning and monitoring for CyberOps.
- Develop policy-driven recommendation memos for senior IT executives, synthesizing federal IT security policy to support strategic decision-making and approvals.
- Manage the CyberOps SharePoint site, establishing it as the centralized hub for cybersecurity knowledge management and streamlining team access to operational and technical resources.

Management and Program Analyst

Jan 2017 – Jun 2022

Supported Supervision, Enforcement and Fair Lending (SEFL) leadership across multiple analyst and program-management roles, leading cross-functional projects that improved operational efficiency and organizational performance. Held Position of Public Trust (PPT).

- Led enterprise-wide programs and projects end-to-end from baseline through close-out spanning process improvement, business intelligence, data analytics, organizational development, and IT application development, delivering measurable gains in efficiency and workforce sustainment.
- Spearheaded the development and standardization of operational playbooks, instituting quality-control processes that improved consistency across teams.
- Built and maintained weekly budget and project dashboards in Tableau, giving senior management real-time visibility to guide resource and staffing decisions.
- Closed Federal Reserve Office of Inspector General findings by assessing supervision examination software (Salesforce) and network access, then designing recurring technology-access review procedures to sustain compliance.
- Managed the internal review and release of SEFL-authored public-facing documents, safeguarding the consistency, accuracy, and transparency of Bureau messaging.

Integrated Systems, Inc. (ISI), Washington, DC

2016

Business Process Reengineering Analyst (Contractor)

Supported Securities and Exchange Commission's (SEC) Office of Information Technology (OIT) SDLC transformation from current to future state, serving as team process SME for business process maps and narratives.

- Analyzed and revised process narrative actors and activities to deploy new and existing SDLC narratives and maps using Visio.
- Created a prototype SDLC SharePoint site to centralize process documentation using SharePoint Designer.
- Developed a training platform in Adobe Captivate to help project managers navigate the new SDLC process.
- Ensured compliance with enterprise architecture directives by converting the future-state SDLC process map into BPMN 2.0 standards using Visio.
- Created UML-based use case models to support requirements development for a new SDLC software automation tool.

Cognizant Technology Services, Arlington, VA

2015

Consultant (Contractor)

Reinforced Assurant's procedural compliance by analyzing, documenting, revising, and deploying internal controls process narratives and maps.

- Facilitated virtual sessions with stakeholders to research, analyze, and revise process narratives and maps (actors, activities, systems, controls) for financial and operational current-state processes using Visio.
- Observed and documented on-site "As-Is" mobile device restoration and disposition processes, ensuring full process transparency.

Cigna, Arlington, VA

2014 – 2015

Business Process Engineer (Contractor)

Supported migration to and integration of a future-state ERP healthcare insurance system from disparate legacy systems by analyzing, developing, and deploying system process flows and change impact evaluations.

- Facilitated virtual sessions with stakeholders to model and deploy "As-Is" and "To-Be" process maps for product, benefit, and client configuration using modified BPMN 2.0 and BlueWorks Live, driving business requirements alignment and establishing level of effort for requirements architecture.
- Analyzed and decomposed business requirements into business rules to support IT system-action determination.

ICMA-RC, Washington, DC

2013

Analyst (Contractor)

Enhanced the IT group's operational efficiency, serving as internal process thought leader in researching best practices and developing, deploying, and maintaining policy, standard, and procedural artifacts across the IT domain.

- Researched best practices to classify, streamline, and deploy group policies, standards, and procedures into a new documentation framework.
- Led client-facing discussions with process owners to develop and deploy user-friendly work process guidelines and templates, standardizing IT enterprise capture of SDLC business, technical, and non-technical requirements.

Deloitte Consulting LLP, Alexandria, VA

2012 – 2013

Business Process Reengineering Analyst (Contractor)

Supported enterprise-wide implementation of the Naval Justice Information System (NJIS), a case-management and workflow-tracking platform, for DON PEO-EIS Sea Warrior Program (PMW 240). Held Secret Clearance.

- Conducted gap analysis of data transferred across naval law enforcement communities to ensure compliance with congressionally mandated reporting requirements.
- Facilitated client-facing working sessions with process owners and stakeholders to assess change impacts and identify information-transfer gaps across actors, activities, systems, and workflows.
- Designed and deployed enterprise-wide “As-Is” and “To-Be” business process maps in BPMN 2.0 using iGrafx, defining current- and future-state process data inputs and outputs.

Leidos (f/k/a Science Applications International Corporation (SAIC)), McLean, VA 2011 – 2012***Senior Business Process Analyst***

Supported the Department of State (DOS) Bureau of Information Resource Management (IRM) Vanguard 2.2.1 transformation from current to future state through the streamlining and consolidation of numerous IT processes. Held Secret Clearance.

- Standardized project process improvement operations by authoring business process templates.
- Reduced performance management data collection cycle time by leveraging a new SharePoint-based tool that automated collection and centralized data storage.
- Monitored subcontractor compliance against monthly contract financial and nonfinancial performance goals, ensuring receipt of maximum contractual incentive revenue.

Booz Allen Hamilton (BAH), McLean, VA

2010 – 2011

Senior Consultant

Supported the Department of the Air Force (DAF) A7C Civil Engineer transformation from current to future state through development, maintenance, and deployment of business process maps and narratives. Held Secret Clearance.

- Constructed, deployed, and revised “As-Is” business process maps in Visio synthesizing interlinked actors, activities, and information to determine capability requirements for a military base housing management IT system.
- Developed and maintained online business process guidebooks and maps in Visio, providing the client with standardized, centrally located operating procedures.
- Managed all IT-related activities between the process project team and IT.

Federal Energy Regulatory Commission (FERC), Washington, DC

2002 – 2010

Energy Industry Analyst

2005 – 2010

Enforced regulatory compliance with financial and operational filing requirements through oversight of a 200-entity oil pipeline and centralized service company portfolio. Held Position of Public Trust (PPT).

- Led design, end-user testing, and deployment of submission software using an SDLC framework, reducing filing costs by a projected 10%.
- Validated and analyzed submitted financial, operational, and transactional data using SQL, Access, and Excel.
- Collaborated in real time with an oil pipeline trade association to ensure proper dissemination of regulatory developments and conveyed association feedback to Commission senior staff.

Organizational Process Consultant (Detailee)

2004 – 2005

Achieved conformity with the Paperwork Reduction Act (PRA) of 1995 by revamping agency-wide current-state information assessment processes and restructuring existing reporting requirements. Held Position of Public Trust (PPT).

- Appraised current-state information requirements through gap analysis, interfacing with internal and external stakeholders to evaluate the effectiveness of collected data and the efficiency of “As-Is” collection and disposition processes.
- Interfaced with federal agencies to formulate best practices for institutionalizing an agency-wide organizational model and processes for assessing current and future information requirements.
- Baselined and mapped core processes in Visio; delivered “To-Be” organizational design and process options in PowerPoint; defined roles and responsibilities for key positions; and deployed an implementation plan to transition the agency from current to future state.

Public Affairs Specialist (Detailee)

2003 – 2004

Redesigned Commission website and resolved public inquiries in a timely manner. Held Position of Public Trust (PPT).

- Benchmarked best practices to improve website usability and visibility.
- Interfaced with internal stakeholders to raise project awareness and solicit input on site design and content.
- Developed content across multiple areas to ensure easy access to information.
- Responded to ~50 public inquiries daily via email and phone, assisting external users in accessing Commission-related information.

Accounting Policy Analyst

2002 – 2003

Researched and assessed accounting issues affecting energy companies, recommending courses of action to anticipate industry direction and practices. Held Position of Public Trust (PPT).

- Developed and delivered training on the potential impact of Sarbanes-Oxley legislation shortly after its introduction.
- Evaluated proposed certificates of public convenience for energy companies seeking to construct U.S. natural gas terminals and pipelines, analyzing financial statements.
- Advised energy companies on Commission accounting regulations.

MILITARY SERVICE

United States Army (USA) - Active Duty, Various Locations, Honorable Discharge, Specialist (E-4)

1992 – 1996

Weapons Locating Radar Operator, Military Occupational Specialty (MOS) 13R

Operated real-time detection and tracking systems to identify, characterize, and report hostile threats under high-stakes operational conditions, directly paralleling the detect-analyze-respond discipline now applied to enterprise cyber threat intelligence, vulnerability management, and incident response processes, as well as the detection, tracking, and threat-mitigation principles central to counter-UAS (C-UAS) operations.

- Determined point of origin and impact of incoming rounds, transmitting accurate targeting data to friendly artillery units and strengthening base security.
- Performed maintenance on radar and associated equipment.

Awarded Army Commendation Medal, Army Achievement Medal, Army Good Conduct Medal and National Defense Service Medal. Grafenwoehr Army Post Soldier of the Month for October 1993.

SCOUTING & YOUTH LEADERSHIP

Scouting America (f/k/a Boy Scouts of America (BSA)) - Troop 162, Wright-Patterson Air Force Base (WPAFB)

Eagle Scout - 1991

Eagle Scout project was the creation and installation of markers for the Wright Brothers Historical Walking Trail at WPAFB. Further information is available [here](#).

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- CompTIA - Security+ (2024)
- Project Management Institute (PMI) - Project Management Professional (PMP) (2018)
- Department of War (DoW) Cybersecurity Workforce Certification Training (CWCT) - Python Institute Certified Associate Python Programmer (PCAP) Exam Prep (2026)
- Cybersecurity and Infrastructure Security Agency (CISA) - Federal Cyber Defense Skilling Academy Pathways: Incident Response (2026), Vulnerability Analysis (2025), and Defensive Cybersecurity (2024); Enterprise Cyber Threat Analysis and Response (2025); Cyber Supply Chain Risk Management (2022)
- Mandiant (now Google Cloud) - Cyber Security Policy and Implementation (2022) and Cyber Intelligence Foundations (2022)

TECHNICAL KNOWLEDGE & FRAMEWORKS

AI & Post-Quantum Governance

- **AI Governance & Risk:** National Institute of Standards and Technology (NIST) AI Risk Management Framework (AI RMF), International Standards Organization/International Electrotechnical Commission (ISO/IEC) 42001 Artificial Intelligence Management System (AIMS), EU AI Act, Office of Management and Budget (OMB) Memorandum M-25-21 (Accelerating Federal Use of AI through Innovation, Governance, and Public Trust), MITRE Adversarial Threat Landscape for Artificial-Intelligence Systems (ATLAS), Open Worldwide Application Security Project (OWASP) Top 10 for Large Language Model (LLM) Applications
- **Post-Quantum Cryptography & Quantum Security:** Executive Order 14412 (Securing the Nation Against Advanced Cryptographic Attacks), Executive Order 14413 (Ushering in the Next Frontier of Quantum Innovation), OMB M-26-15 (Execution of the Migration to Post-Quantum Cryptography), NIST SP 800-208 (Recommendation for Stateful Hash-Based Signature Schemes)

Governance, Risk & Compliance

- **GRC Frameworks:** NIST Cybersecurity Framework (CSF), NIST Special Publication (SP) 800-37 (RMF), NIST SP 800-53 (incl. IA and SA/SR Control Families), NIST SP 800-53A (Assessment Procedures), NIST SP 800-171 (Protecting Controlled Unclassified Information (CUI) in Nonfederal Systems and Organizations), NIST SP 800-30 (Risk Assessment), NIST SP 800-39 (Risk Management), NIST SP 800-128 (Security-Focused Configuration Management), NIST Federal Information Processing Standard (FIPS) 199 (Standards for Security Categorization of Federal Information and Information Systems), NIST FIPS 200 (Minimum Security Requirements for Federal Information and Information Systems), ISO/IEC 27001/27002 (Information Security, Cybersecurity and Privacy Protection - Information Security Management Systems - Requirements/ Information Security, Cybersecurity and Privacy Protection - Information Security Controls), ISO/IEC 20000 (IT Service Management), Information Systems Audit and Control Association (ISACA) Control Objectives for Information and Related Technologies (COBIT), Committee of Sponsoring Organizations of the Treadway Commission (COSO) Internal Control-Integrated Framework, Center for

Internet Security (CIS) Critical Security Controls (CIS 18), NIST SP 1800-5 (IT Asset Management), NIST SP 1800-27 (Securing Property Management Systems)

- **Federal Cybersecurity Directives:** Cybersecurity and Infrastructure Security Agency (CISA) Binding Operational Directives (BOD), including BOD 26-04 (Prioritizing Security Updates Based on Risk), BOD 26-02 (Mitigating Risk from End-of-Support Edge Devices), BOD 25-01 (Implementing Secure Practices for Cloud Services), BOD 23-01 (Implementation Guidance for Improving Asset Visibility and Vulnerability Detection on Federal Networks), and BOD 20-01 (Develop and Publish a Vulnerability Disclosure Policy); CISA Emergency Directives (ED); National Security Presidential Memorandum-12 (NSPM-12, Cybersecurity of National Security Systems); National Security Memorandum-22 (NSM-22, Critical Infrastructure Security and Resilience)
- **Compliance & Certification:** System and Organization Controls (SOC) 2 (Type I/II), Cybersecurity Maturity Model Certification (CMMC) 2.0, Defense Federal Acquisition Regulation Supplement (DFARS) 252.204-7012, NIST SP 800-171A, Payment Card Industry Data Security Standard (PCI DSS), US Federal Information Security Management Act (FISMA), European Union (EU) General Data Protection Regulation (GDPR), Executive Order 14028 (Improving the Nation's Cybersecurity), Executive Order 14306 (Sustaining Select Efforts to Strengthen the Nation's Cybersecurity and Amending Executive Order 13694 and Executive Order 14144), Executive Order 14409 (Promoting Advanced AI Innovation and Security)

Security Operations & Architecture

- **Threat & Vulnerability Analysis:** MITRE Adversarial Tactics, Techniques, and Common Knowledge (ATT&CK), Lockheed Martin Cyber Kill Chain, MITRE Common Vulnerabilities and Exposures/Common Weakness Enumeration (CVE/CWE), NIST National Vulnerability Database (NVD), Forum of Incident Response and Security Teams (FIRST) Common Vulnerability Scoring System (CVSS) v4, FIRST Exploit Prediction Scoring System (EPSS), CISA Stakeholder-Specific Vulnerability Categorization (SSVC), MITRE Common Attack Pattern Enumeration and Classification (CAPEC), CISA Known Exploited Vulnerabilities (KEV) Catalog, Structured Threat Information eXpression/Trusted Automated eXchange of Intelligence Information (STIX)/TAXII Threat Intelligence Sharing, OWASP Risk Rating Methodology, Factor Analysis of Information Risk (FAIR), Breach and Attack Simulation (BAS), NIST SP 800-40 Rev. 4 (Guide to Enterprise Patch Management Planning), NIST SP 1800-31 (Improving Enterprise Patching for General IT Systems)
- **Incident Response Frameworks & Authorities:** NIST SP 800-61 Rev. 3 (Incident Response Recommendations and Considerations for Cybersecurity Risk Management), NIST SP 800-86 (Digital Forensics), ISO/IEC 27035 (Information Technology - Information Security Incident Management), SANS Incident Response Framework, CISA National Cyber Incident Scoring System (NCISS), CISA Federal Incident Notification Guidelines, Presidential Policy Directive 41 (PPD-41) (US Cyber Incident Coordination), CIRCIA (Cyber Incident Reporting for Critical Infrastructure Act of 2022), OMB M-26-14 (Ensuring Effective and Efficient Agency Logging and Network Visibility to Defend Against Evolving Cyber Threats), NIST SP 1800-25 (Data Integrity: Identifying and Protecting Assets Against Ransomware and Other Destructive Events), NIST SP 1800-26 (Data Integrity: Detecting and Responding to Ransomware and Other Destructive Events), NIST SP 1800-28 (Data Confidentiality: Identifying and Protecting Assets Against Data Breaches), NIST SP 1800-29 (Data Confidentiality: Detect, Respond to, and Recover from Data Breaches)
- **Cloud Security & FedRAMP:** NIST SP 800-144/145 (Cloud Computing Security/Definition), OMB M-24-15 (Modernizing Federal Risk and Authorization Management Program (FedRAMP)), FedRAMP Rev5 & FedRAMP 20x/Consolidated Rules 2026 (CR26), FedRAMP Authorization Boundaries, Cloud Security Alliance (CSA) Cloud Controls Matrix (CCM), NIST SP 800-201 (NIST Cloud Computing Forensic Reference Architecture), NIST SP 800-209

(Security Guidelines for Storage Infrastructure), NIST SP 800-210 (General Access Control Guidance for Cloud Systems)

- **Secure Software Development:** OMB M-26-05 (Adopting a Risk-Based Approach to Software and Hardware Security), NIST SP 800-218 (Secure Software Development Framework/SSDF), STRIDE/PASTA/DREAD Threat Modeling Methodologies, OWASP Software Assurance Maturity Model (SAMM), OWASP Top 10 (Web Application), OWASP Web Security Testing Guide (WSTG), OWASP Application Security Verification Standard (ASVS), OWASP Mobile Application Security Verification Standard (MASVS)/Mobile Security Testing Guide (MSTG), Penetration Testing Execution Standard (PTES), Software Supply Chain Risk Management (Software Bill of Materials (SBOM)), ISO/IEC 15288 (Systems Engineering Life Cycle), ISO/IEC 12207 (Software Life Cycle), ISO/IEC/IEEE 90003 (Software Engineering - Guidelines for the Application of ISO 9001 to Computer Software), DoD DevSecOps Reference Design, NIST SP 800-204 (Security Strategies for Microservices-Based Application Systems), NIST SP 800-204A (Building Secure Microservices-Based Applications Using Service-Mesh Architecture), NIST SP 800-204B (Attribute-Based Access Control for Microservices-Based Applications Using a Service Mesh), NIST SP 800-204C (Implementation of DevSecOps for a Microservices-Based Application with Service Mesh), NIST SP 800-204D (Strategies for the Integration of Software Supply Chain Security in DevSecOps CI/CD Pipelines)
- **Identity & Access Management:** NIST SP 800-63 (Digital Identity Guidelines), NIST SP 800-207A (Zero Trust IAM Framework), NIST SP 800-205 (Attribute Considerations for Access Control Systems), Multi-Factor Authentication (MFA), Privileged Access Management (PAM), Federated Identity/SSO, OAuth 2.0, SAML, OpenID Connect
- **Zero Trust Frameworks & Strategy:** CISA Zero Trust Maturity Model 2.0, NIST SP 800-207 (Zero Trust Architecture), OMB M-22-09 (Federal Zero Trust Strategy), NIST SP 800-160 (Systems Security Engineering), NIST SP 1800-35 (Implementing a Zero Trust Architecture)
- **IoT & Firmware Security:** NIST SP 800-193 (Platform Firmware Resiliency Guidelines), NIST SP 800-213 (IoT Device Cybersecurity Guidance for the Federal Government: Establishing IoT Device Cybersecurity Requirements), NIST SP 800-213A (IoT Device Cybersecurity Requirement Catalog), NIST SP 1800-34 (Validating the Integrity of Computing Devices), NIST SP 1800-36 (Trusted Internet of Things (IoT) Device Network-Layer Onboarding and Lifecycle Management)

Specialized Regulatory & Industry Compliance

- **Unmanned Aircraft Systems (UAS) & Counter-UAS Security:** Federal Aviation Administration (FAA) Part 107 (Unmanned Aircraft Systems Operations), National Defense Authorization Act (NDAA) Section 848 & Blue UAS Framework (Cleared UAS Supply Chain/Cybersecurity Vetting), American Security Drone Act, OMB Memorandum M-26-02 (Ensuring Government Use of Secure Unmanned Aircraft Systems and Supporting United States Producers); Counter-UAS (C-UAS): Preventing Emerging Threats Act of 2018, 6 U.S.C. § 124n (Protection of Certain Facilities and Assets from Unmanned Aircraft), 49 U.S.C. § 44810 (FAA C-UAS Authority), Department of Defense (DoD) Counter-Small Unmanned Aircraft Systems (C-sUAS) Strategy, Joint Counter-small Unmanned Aircraft Systems Office (JCO) Coordination, Radio Frequency (RF)/Global Navigation Satellite System (GNSS) Detection & Mitigation Principles
- **Aviation & Maritime Sector Security:** AS9100 (Aerospace Quality Management Systems), Radio Technical Commission for Aeronautics (RTCA) DO-356A (Airworthiness Security Methods and Considerations), Maritime Transportation Security Act (MTSA) of 2002, 33 CFR Subchapter H (Maritime Security), 33 CFR Parts 101-106 (Coast Guard Cybersecurity in the Marine Transportation System Final Rule, 2024), International Maritime Organization (IMO) Resolution MSC.428(98) (Maritime Cyber Risk Management in Safety Management Systems), IMO MSC-FAL.1/Circ.3 (Guidelines on Maritime Cyber Risk Management), BIMCO Guidelines on Cyber Security Onboard Ships, 14 CFR Part 25 Special Conditions for Aircraft

Cybersecurity, RTCA DO-326A/EUROCAE ED-202A (Airworthiness Security Process Specification), European Union Aviation Safety Agency (EASA) Part-IS (Information Security Regulation), Transportation Security Administration (TSA) Security Directives (Aviation Cybersecurity, SD 1542-21-01 Series), International Civil Aviation Organization (ICAO) Annex 17 (Aviation Security) and Aviation Cybersecurity Strategy; Homeland Security Act of 2002, Aviation and Transportation Security Act (ATSA) of 2001, and Cybersecurity Information Sharing Act (CISA) of 2015 as cross-cutting authorities underpinning TSA and DHS Transportation Systems Sector cybersecurity oversight

- **Energy & Water Infrastructure Security:** ISO 29001 (Petroleum, Petrochemical and Natural Gas Industries Quality Management Systems), US Environmental Protection Agency (EPA) Water and Wastewater Systems (WWS) Sector Risk Management Agency (SRMA), American Water Works Association (AWWA) G430 (Security Practices for Operation and Management) & J100 (Risk and Resilience Management), America's Water Infrastructure Act (AWIA) of 2018 (Risk & Resilience Assessments and Emergency Response Plans), North American Electric Reliability Corporation (NERC) Critical Infrastructure Protection (CIP) Standards, Transportation Security Administration (TSA) Security Directive Pipeline-2021 Series (Pipeline Cybersecurity Mitigation Actions, Contingency Planning, and Testing), Presidential Policy Directive 21 (PPD-21, Critical Infrastructure Security and Resilience), Executive Order 13920 (Securing the United States Bulk-Power System), NIST SP 1800-2 (Identity and Access Management for Electric Utilities), NIST SP 1800-7 (Situational Awareness for Electric Utilities), NIST SP 1800-23 (Energy Sector Asset Management: For Electric Utilities, Oil & Gas Industry), NIST SP 1800-32 (Securing Distributed Energy Resources: An Example of Industrial Internet of Things Cybersecurity), NIST SP 1800-45 (Cybersecurity for the Water and Wastewater Sector: Build Architecture (Operational Technology Remote Access))
- **Manufacturing & Industrial Control Security:** International Electrotechnical Commission (IEC) 62443 (Industrial Automation and Control Systems (IACS) Cybersecurity), NIST Special Publication (SP) 800-82 (Guide to Operational Technology (OT) Security), CISA Cross-Sector Cybersecurity Performance Goals (CPGs), MITRE ATT&CK for ICS, NIST SP 1800-10 (Protecting Information and System Integrity in Industrial Control System Environments: Cybersecurity for the Manufacturing Sector)
- **Automotive Sector Security:** UN Regulation No. 155 (UN R155, Cybersecurity Management System), UN Regulation No. 156 (UN R156, Software Update Management System), ISO/SAE 21434 (Road Vehicles - Cybersecurity Engineering), ISO 24089 (Software Update Engineering), ISO/TS 16949/IATF 16949 (Automotive Quality Management Systems)
- **Healthcare & Medical Device Security:** U.S. Food and Drug Administration (FDA) Cybersecurity in Medical Devices Guidance (Quality Management System Considerations and Content of Premarket Submissions), FDA Quality System Regulation (QSR)/21 CFR Part 820, ISO 13485 (Medical Devices Quality Management Systems), Health Insurance Portability and Accountability Act (HIPAA) Security Rule, Health Information Technology for Economic and Clinical Health (HITECH) Act, Health Information Trust Alliance (HITRUST) Common Security Framework (CSF), ISO/IEC 27799 (Health Informatics - Information Security Management), HHS 405(d) Health Industry Cybersecurity Practices (HICP), NIST Special Publication (SP) 800-66 (HIPAA Security Rule Implementation Guide), UK National Health Service (NHS) Data Security and Protection Toolkit (DSPT), EU Network and Information Security Directive 2 (NIS2), NIST SP 1800-1 (Securing Electronic Health Records on Mobile Devices), NIST SP 1800-8 (Securing Wireless Infusion Pumps in Healthcare Delivery Organizations), NIST SP 1800-24 (Securing Picture Archiving and Communication System (PACS): Cybersecurity for the Healthcare Sector), NIST SP 1800-30 (Securing Telehealth Remote Patient Monitoring Ecosystem)
- **FERC Regulatory Framework:** 18 CFR Part 35 (Filing of Rate Schedules), 18 CFR Part 157 (Applications for Certificates of Public Convenience and Necessity), 18 CFR Part 357 (Reports of Pipe Line Companies), 18 CFR Part 366 (Filing Requirements for Public Utility Holding Companies)

Program Management, Data Policy & Acquisition

- **Process, Quality & Program Management:** Capability Maturity Model Integration (CMMI), CMMI AI Maturity (AIM), Six Sigma/Lean, ISO 9001 (Quality Management Systems), Object Management Group (OMG) Business Process Model and Notation (BPMN) 2.0, OMG Unified Modeling Language (UML), PMI Project Management Body of Knowledge (PMBOK), ISO 21500/21502/21503/21504/21505 (Project, Program and Portfolio Management), NIST SP 800-181 (National Initiative for Cybersecurity Education (NICE) Cybersecurity Workforce Framework), U.S. Government Accountability Office (GAO) Business Process Reengineering Assessment Guide - Version 3, 10 U.S.C. § 2222 (Defense Business Systems: Business Process Reengineering; Enterprise Architecture; Management), DoD Instruction (DoDI) 5000.75 (Business Systems Requirements and Acquisition/Business Capability Acquisition Cycle (BCAC)), Department of War (DoW) Business Process Reengineering Center (BPRC), DoD Architecture Framework (DoDAF) 2.02, Business Enterprise Architecture (BEA)
- **Data Policy & Privacy:** Privacy Act of 1974, E-Government Act of 2002 (Privacy Impact Assessments), Freedom of Information Act (FOIA); OMB M-19-23 (Federal Data Strategy), OMB Circular A-130 (Data as a Strategic Asset), Data Management Association (DAMA)-DMBOK (Data Management Body of Knowledge), ISO 8000 (Data Quality), ISO/IEC 11179 (Metadata Registries)
- **Acquisition & Records Management:** Federal Acquisition Regulation (FAR), Paperwork Reduction Act (PRA) of 1995, Federal Records Act, National Archives and Records Administration (NARA) Federal Records Retention Requirements

TECHNICAL SKILLS & TOOLS

Security Tools & Technologies

- **Penetration Testing:** Kali Linux, Bugcrowd, Burp Suite, Metasploit, OWASP ZAP, Autopsy, Netcat, Armitage, SQLMap, Recon-ng, Hashcat, Hydra, Ettercap, Zenmap, Hping3, NetExec, SEToolkit, Donut, Aircrack-ng
- **Detection, Monitoring & Response:** Qualys & OpenVAS (Vulnerability Management), Anomali ThreatStream (Threat Intelligence), CrowdStrike Falcon (Endpoint Detection and Response (EDR)), urlscan.io (URL/Phishing Analysis), Splunk & Cribl (Security Information and Event Management (SIEM)/Data Pipeline)
- **Networking & Protocols:** Open Systems Interconnection (OSI) Model, Transmission Control Protocol/Internet Protocol (TCP/IP) Model, Wireshark, Nmap, Snort, pfSense (Firewall/Network Security)

Development & Automation

- **AI & Process Automation:** Claude, Claude Cowork, Microsoft Power Automate
- **Languages & Scripting:** Python, SQL, PowerShell, Visual Basic for Applications (VBA), Git
- **Development Tools & Cloud:** Visual Studio Code, Visual Studio, Jupyter Notebook, GitHub, Amazon Web Services (AWS), Docker, Kubernetes

Enterprise Platforms

- **IT Service, Project & Business Process Management:** ServiceNow, Jira, Microsoft Project, Microsoft Visio, IBM BlueWorks Live, iGrafx
- **Data Analytics & Business Intelligence (BI):** Tableau, Microsoft Excel, Microsoft Access
- **Collaboration, Content & Business Applications (CRM, HCM, Experience):** Microsoft SharePoint, SharePoint Designer, Salesforce, SAP SuccessFactors, Qualtrics